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$$\begin{aligned}\lim_{x \rightarrow -\frac{3}{2}} \frac{2x+3}{4x^2+12x+9} &= \frac{0}{0} \\ \lim_{x \rightarrow -\frac{3}{2}} \frac{2x+3}{(2x+3)^2} &= \frac{1}{2x+3} = \frac{1}{0} \\ \left\{ \begin{array}{l} \lim_{x \rightarrow -\frac{3}{2}-} \frac{1}{2x+3} = -\infty \\ \lim_{x \rightarrow -\frac{3}{2}+} \frac{1}{2x+3} = +\infty \end{array} \right. \\ \Rightarrow \lim_{x \rightarrow -\frac{3}{2}} \frac{2x+3}{4x^2+12x+9} &\text{ bestaat niet}\end{aligned}$$

References